Title

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Abstract

Maximum 250 words.
Keywords: keyword1, keyword2, keyword3.

1 Introduction

1.1 Application setup

This work proposes a controlled numerical investigation of sparse linear systems arising from the semi-implicit discretization of the two-dimensional linearized shallow water equations. The objective is not to develop a full oceanographic simulation, but to use this model as a test case for studying the conditioning of linear systems and the convergence of iterative solvers.

A semi-implicit finite-difference formulation will be adopted, following the approaches discussed by Casulli [3] and Bonaventura and Rosatti [2]. Such formulations lead to sparse linear systems that must be solved at each time step.

The experiments will consider a rectangular computational domain, starting with a constant reference water depth. A second set of experiments will introduce controlled spatial variations in the depth coefficient. The investigation will examine how grid refinement and increasing coefficient contrast affect the resulting matrix, the number of iterations required for convergence, and the computational cost of the solution process.

As a methodological extension, the work will compare the Conjugate Gradient method without preconditioning with preconditioned variants using Jacobi and incomplete Cholesky preconditioners. The applicability of Conjugate Gradient will first be verified by checking the symmetry and positive definiteness of the discrete system. This comparison is motivated by

the established literature on iterative methods for sparse linear systems and preconditioning techniques [4, 1].

The evaluation will consider iteration counts, computational time, relative residual norms, and estimates of the condition number, where computationally feasible. Solver tolerances and other experimental settings will be controlled to enable meaningful comparisons across grid resolutions and coefficient configurations.

The main research question is how grid refinement and coefficient heterogeneity affect iterative convergence, and to what extent incomplete Cholesky preconditioning can improve solver performance relative to the unpreconditioned and Jacobi-preconditioned methods.

2 Methods and techniques

Describe the methods and techniques used in the work. The work must be an application of the concepts learned in the course, and must involve at the minimum one of the following topics: linear systems, optimization, eigenvalues and eigenvectors, advanced integration.

3 Results

Specific tests and results.

4 Conclusions

Present the conclusions of the work.

References

[1] Michele Benzi. Preconditioning techniques for large linear systems: a survey. *Journal of Computational Physics*, 182(2):418–477, 2002. doi: 10.1006/jcph.2002.7176. URL https://www.sciencedirect.com/science/article/pii/S0021999102971767.

[2] Luca Bonaventura and Giorgio Rosatti. A cascadic conjugate gradient algorithm for mass conservative, semi-implicit discretization of the shallow water equations on locally refined structured grids. *International Journal for Numerical Methods in Fluids*, 40(1-2):217–230, 2002. doi: 10.1002/fld.274. URL https://doi.org/10.1002/fld.274.

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